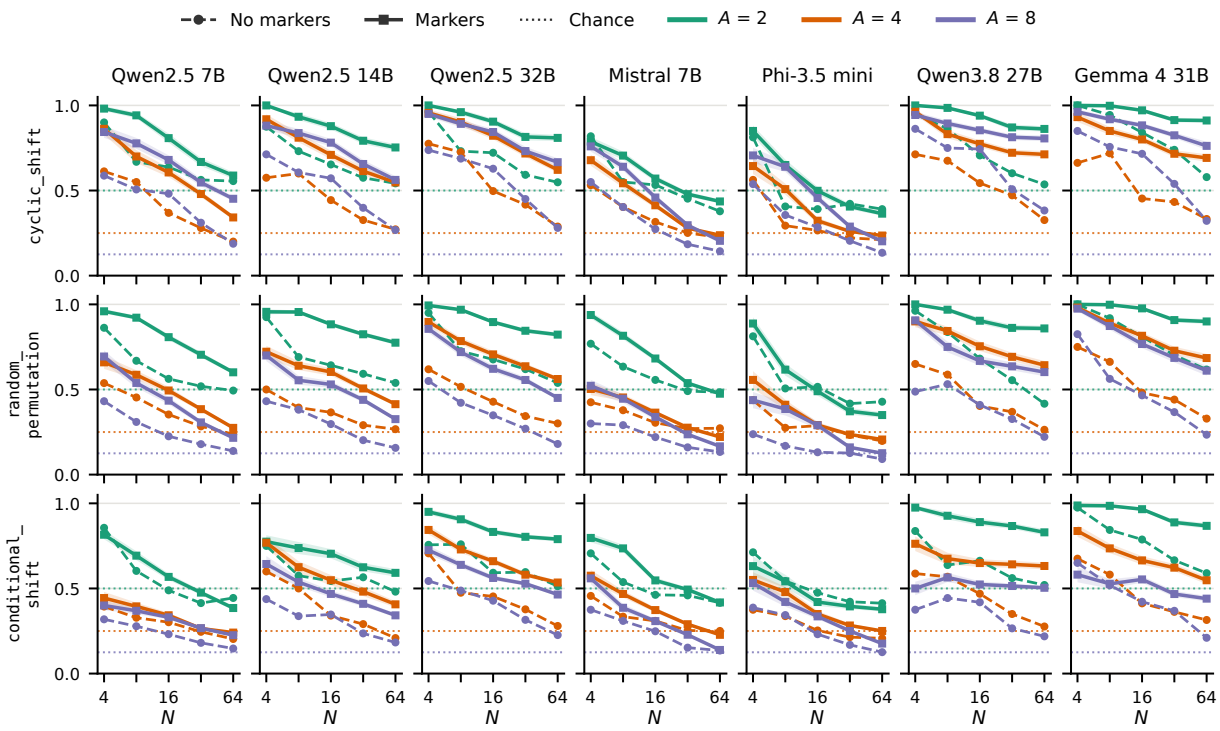



Figure 7: Position-local tasks on contiguous letters, by model and length. Token accuracy against sequence length $N$ (4 to 64). Dashed: no markers; solid: markers (mean over spacings $k \in \{1, 2, 4, 8, 16\}$ with $k < N$); colors: alphabet size; dotted: chance; bands: $\pm 1$ standard error.

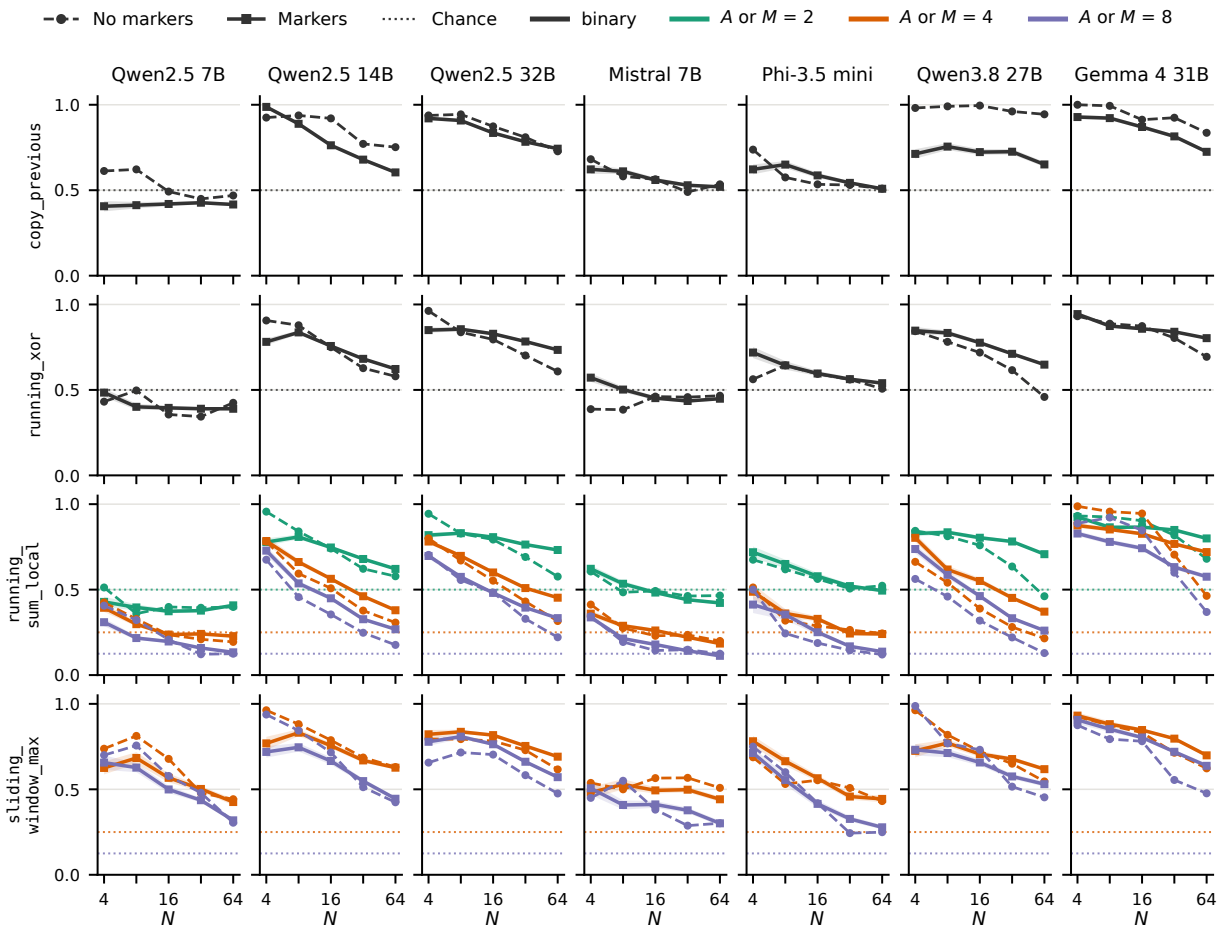



Figure 8: Dependency tasks (digits), by model and length; same conventions as Figure 7. Colors: modulus (`running_sum_local`) or alphabet size (`sliding_window_max`); black: binary. Marker runs average over all positions, so they mix the post-marker loss with the within-segment gain (Section 6).

**Out-of-alphabet answers.** On the digit tasks, Qwen2.5-7B often writes a letter (usually `t`, sometimes `o`) in place of `0`, in 34% to 39% of outputs, with or without markers. On spaced letters, several models step outside the alphabet (for example `I` after `H` instead of wrapping to `A`). Such characters are scored as errors.

## D Per-position accuracy

Table 6 gives the per-position means plotted in Figure 3, and Table 7 breaks them down by distance from the marker. Table 8 repeats the dependency-task comparison under alignment-robust scoring, and Figure 9 shows the length profile of the tokenization controls for each task. "No markers" is the no-marker run's accuracy at the same position indices. Means are over the tasks in each block; per-task values are in `results_clean/_summary/per_position_values.csv`.

Table 6: Per-position accuracy by model ($N \leq 64$, $k \in \{2, 4, 8, 16\}$), mean over the tasks in each block.

| | Within a segment | | | Right after a marker | | |
|---|---|---|---|---|---|---|
| Model | No markers | Markers | Δ | No markers | Markers | Δ |
| *Position-local tasks, letters* | | | | | | |
| Qwen2.5 7B | 0.38 | 0.45 | +0.07 | 0.37 | 0.51 | +0.14 |
| Qwen2.5 14B | 0.43 | 0.55 | +0.11 | 0.40 | 0.63 | +0.23 |
| Qwen2.5 32B | 0.49 | 0.65 | +0.16 | 0.45 | 0.79 | +0.34 |
| Mistral 7B | 0.35 | 0.35 | 0.00 | 0.30 | 0.39 | +0.09 |
| Phi-3.5 mini | 0.31 | 0.31 | 0.00 | 0.28 | 0.35 | +0.07 |
| Qwen3.8 27B | 0.50 | 0.69 | +0.20 | 0.47 | 0.81 | +0.34 |
| Gemma 4 31B | 0.56 | 0.72 | +0.16 | 0.51 | 0.80 | +0.29 |
| Gemini 3.5 Flash | 0.57 | 0.74 | +0.17 | 0.52 | 0.83 | +0.32 |
| *Position-local tasks, spaced letters* | | | | | | |
| Qwen2.5 7B | 0.54 | 0.55 | +0.01 | 0.52 | 0.52 | 0.00 |
| Qwen2.5 14B | 0.69 | 0.80 | +0.11 | 0.67 | 0.82 | +0.15 |
| Qwen2.5 32B | 0.79 | 0.93 | +0.14 | 0.79 | 0.92 | +0.14 |
| Mistral 7B | 0.40 | 0.43 | +0.03 | 0.35 | 0.45 | +0.09 |
| Phi-3.5 mini | 0.35 | 0.35 | −0.01 | 0.33 | 0.34 | +0.01 |
| Qwen3.8 27B | 0.83 | 0.92 | +0.09 | 0.84 | 0.92 | +0.08 |
| Gemma 4 31B | 0.88 | 0.94 | +0.06 | 0.89 | 0.94 | +0.05 |
| *Position-local tasks, digits* | | | | | | |
| Qwen2.5 7B | 0.35 | 0.38 | +0.04 | 0.30 | 0.35 | +0.05 |
| Qwen2.5 14B | 0.77 | 0.85 | +0.08 | 0.77 | 0.87 | +0.09 |
| Qwen2.5 32B | 0.83 | 0.94 | +0.11 | 0.83 | 0.93 | +0.09 |
| Mistral 7B | 0.43 | 0.38 | −0.04 | 0.41 | 0.39 | −0.02 |
| Phi-3.5 mini | 0.48 | 0.55 | +0.07 | 0.44 | 0.56 | +0.12 |
| Qwen3.8 27B | 0.85 | 0.94 | +0.09 | 0.85 | 0.94 | +0.09 |
| Gemma 4 31B | 0.92 | 0.99 | +0.07 | 0.92 | 0.98 | +0.06 |
| *Dependency tasks* | | | | | | |
| Qwen2.5 7B | 0.43 | 0.41 | −0.02 | 0.42 | 0.36 | −0.06 |
| Qwen2.5 14B | 0.68 | 0.68 | 0.00 | 0.64 | 0.49 | −0.15 |
| Qwen2.5 32B | 0.70 | 0.77 | +0.07 | 0.66 | 0.52 | −0.15 |
| Mistral 7B | 0.44 | 0.43 | 0.00 | 0.43 | 0.38 | −0.04 |
| Phi-3.5 mini | 0.49 | 0.48 | 0.00 | 0.45 | 0.42 | −0.03 |
| Qwen3.8 27B | 0.68 | 0.70 | +0.02 | 0.63 | 0.44 | −0.19 |
| Gemma 4 31B | 0.79 | 0.85 | +0.05 | 0.78 | 0.52 | −0.26 |
| Gemini 3.5 Flash | 0.92 | 0.97 | +0.05 | 0.92 | 0.49 | −0.43 |
| *Dependency tasks, reset variants* | | | | | | |
| Qwen2.5 7B | 0.43 | 0.43 | 0.00 | 0.41 | 0.58 | +0.17 |
| Qwen2.5 32B | 0.69 | 0.79 | +0.10 | 0.65 | 0.92 | +0.27 |
| Qwen3.8 27B | 0.66 | 0.68 | +0.03 | 0.63 | 0.98 | +0.36 |
| Gemma 4 31B | 0.77 | 0.85 | +0.08 | 0.75 | 0.97 | +0.22 |

Table 7: Accuracy by distance after the marker at $k = 16$ ($N \in \{32, 64\}$): 0 is the position right after the marker. Mean over the tasks in each block.

| | | Positions after the marker | | | | |
|---|---|---|---|---|---|---|
| Tasks | Model | 0 | 1 | 2–3 | 4–7 | 8–15 |
| Position-local, letters | Qwen2.5 7B | 0.32 | 0.38 | 0.35 | 0.29 | 0.26 |
| | Qwen2.5 14B | 0.45 | 0.47 | 0.41 | 0.37 | 0.34 |
| | Qwen2.5 32B | 0.60 | 0.57 | 0.48 | 0.45 | 0.41 |
| | Mistral 7B | 0.28 | 0.34 | 0.29 | 0.26 | 0.25 |
| | Phi-3.5 mini | 0.30 | 0.28 | 0.28 | 0.28 | 0.23 |
| | Qwen3.8 27B | 0.68 | 0.67 | 0.59 | 0.51 | 0.47 |
| | Gemma 4 31B | 0.52 | 0.62 | 0.55 | 0.51 | 0.43 |
| | Gemini 3.5 Flash | 0.74 | 0.69 | 0.65 | 0.51 | 0.46 |
| Position-local, spaced | Qwen2.5 7B | 0.30 | 0.46 | 0.47 | 0.43 | 0.35 |
| | Qwen2.5 14B | 0.59 | 0.67 | 0.68 | 0.66 | 0.59 |
| | Qwen2.5 32B | 0.86 | 0.90 | 0.89 | 0.88 | 0.85 |
| | Mistral 7B | 0.31 | 0.39 | 0.37 | 0.27 | 0.24 |
| | Phi-3.5 mini | 0.25 | 0.37 | 0.33 | 0.25 | 0.21 |
| | Qwen3.8 27B | 0.89 | 0.88 | 0.91 | 0.91 | 0.90 |
| | Gemma 4 31B | 0.93 | 0.94 | 0.95 | 0.93 | 0.94 |
| Position-local, digits | Qwen2.5 7B | 0.28 | 0.38 | 0.32 | 0.30 | 0.25 |
| | Qwen2.5 14B | 0.83 | 0.85 | 0.86 | 0.81 | 0.76 |
| | Qwen2.5 32B | 0.90 | 0.93 | 0.94 | 0.93 | 0.90 |
| | Mistral 7B | 0.31 | 0.38 | 0.36 | 0.31 | 0.26 |
| | Phi-3.5 mini | 0.38 | 0.53 | 0.52 | 0.46 | 0.37 |
| | Qwen3.8 27B | 0.90 | 0.89 | 0.91 | 0.91 | 0.90 |
| | Gemma 4 31B | 0.94 | 0.98 | 0.99 | 0.98 | 0.98 |
| Dependency | Qwen2.5 7B | 0.36 | 0.41 | 0.42 | 0.42 | 0.36 |
| | Qwen2.5 14B | 0.45 | 0.75 | 0.68 | 0.60 | 0.53 |
| | Qwen2.5 32B | 0.49 | 0.85 | 0.79 | 0.71 | 0.63 |
| | Mistral 7B | 0.41 | 0.42 | 0.45 | 0.42 | 0.41 |
| | Phi-3.5 mini | 0.40 | 0.47 | 0.44 | 0.40 | 0.38 |
| | Qwen3.8 27B | 0.46 | 0.70 | 0.71 | 0.65 | 0.58 |
| | Gemma 4 31B | 0.49 | 0.82 | 0.83 | 0.81 | 0.75 |
| | Gemini 3.5 Flash | 0.57 | 1.00 | 1.00 | 1.00 | 0.97 |

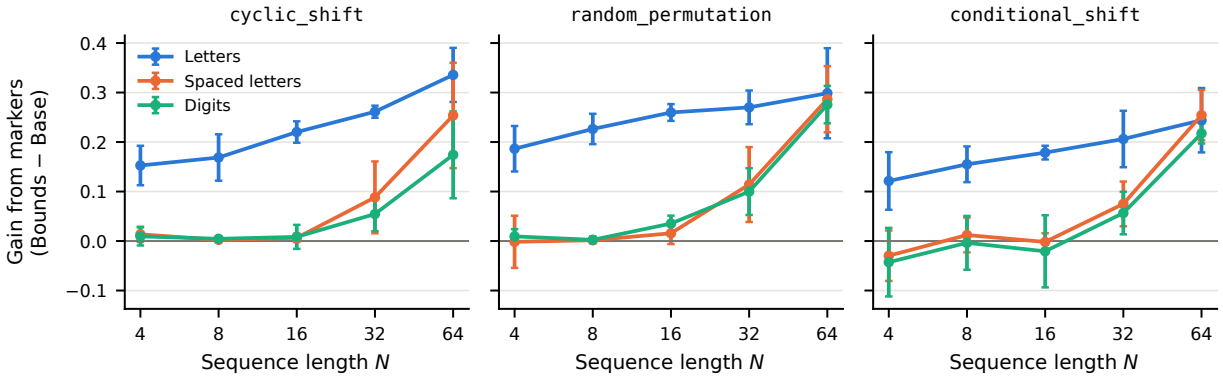



Figure 9: The length profile of Figure 4a, task by task (mean over Qwen2.5-14B and 32B, Qwen3.8-27B, and Gemma-4-31B; whiskers are 95% intervals over models). All three tasks show the same pattern.

Table 8: Per-position gains on dependency tasks (marker run minus the no-marker run at the same positions) under index-by-index scoring and under edit-distance alignment, which repairs dropped or inserted characters. $N \leq 64$, $k \in \{2, 4, 8, 16\}$.

| | | Right after a marker | | Within a segment | |
|---|---|---|---|---|---|
| Task | Model | Index | Aligned | Index | Aligned |
| `copy_previous` | Qwen2.5 7B | −0.10 | −0.09 | −0.07 | −0.06 |
| | Qwen2.5 14B | −0.21 | −0.15 | −0.10 | −0.05 |
| | Qwen2.5 32B | −0.15 | −0.17 | −0.02 | −0.02 |
| | Mistral 7B | −0.05 | −0.01 | 0.00 | +0.01 |
| | Phi-3.5 mini | −0.04 | −0.07 | −0.01 | −0.04 |
| | Qwen3.8 27B | −0.45 | −0.20 | −0.24 | −0.05 |
| | Gemma 4 31B | −0.29 | −0.25 | −0.07 | −0.04 |
| `running_xor` | Qwen2.5 7B | +0.02 | −0.02 | +0.01 | 0.00 |
| | Qwen2.5 14B | −0.14 | −0.13 | +0.05 | +0.01 |
| | Qwen2.5 32B | −0.19 | −0.24 | +0.11 | +0.03 |
| | Mistral 7B | −0.04 | −0.03 | +0.02 | −0.03 |
| | Phi-3.5 mini | −0.05 | −0.02 | +0.03 | +0.03 |
| | Qwen3.8 27B | −0.06 | −0.08 | +0.13 | +0.06 |
| | Gemma 4 31B | −0.26 | −0.29 | +0.09 | +0.04 |
| `running_sum_local` | Qwen2.5 7B | −0.03 | −0.04 | −0.01 | −0.02 |
| | Qwen2.5 14B | −0.08 | −0.12 | +0.07 | +0.03 |
| | Qwen2.5 32B | −0.16 | −0.22 | +0.11 | +0.05 |
| | Mistral 7B | −0.01 | −0.04 | 0.00 | −0.01 |
| | Phi-3.5 mini | −0.03 | 0.00 | −0.02 | −0.03 |
| | Qwen3.8 27B | −0.06 | −0.13 | +0.14 | +0.09 |
| | Gemma 4 31B | −0.41 | −0.46 | +0.11 | +0.07 |
| `sliding_window_max` | Qwen2.5 7B | −0.11 | −0.06 | −0.02 | −0.02 |
| | Qwen2.5 14B | −0.16 | −0.10 | −0.03 | −0.01 |
| | Qwen2.5 32B | −0.09 | −0.07 | +0.08 | +0.07 |
| | Mistral 7B | −0.07 | 0.00 | −0.02 | 0.00 |
| | Phi-3.5 mini | +0.02 | −0.03 | −0.02 | −0.03 |
| | Qwen3.8 27B | −0.18 | −0.16 | +0.05 | +0.04 |
| | Gemma 4 31B | −0.07 | −0.08 | +0.08 | +0.02 |

## E Reset variants, pooled

Table 9 gives the pooled effect of markers on each dependency task in its original and reset forms; the per-position comparison is in Section 6.1.

Table 9: Reset variants, pooled over positions: TAR Base → Bounds, Δ, and $|z|$, pooled as in Figure 5 (no-marker mean length-matched). Base differs between the original and reset columns because the reset prompt alone shifts no-marker accuracy; Section 6.1 therefore compares marker runs position by position. Originals use 20 to 80 trials per cell (Table 5); reset variants 20, except 80 at Qwen2.5-7B.

| | | Original (dependency intact) | | Reset (dependency severed) | |
|---|---|---|---|---|---|
| Task | Model | Base→Bounds | Δ ($\|z\|$) | Base→Bounds | Δ ($\|z\|$) |
| `running_xor` | 7B | 0.401→0.403 | +0.002 (0.1) | 0.450→0.482 | +0.032 (3.4) |
| | 32B | 0.746→0.798 | +0.053 (6.2) | 0.715→0.878 | +0.163 (11.9) |
| | Qwen3.8 27B | 0.646→0.742 | +0.096 (9.5) | 0.645→0.855 | +0.210 (14.4) |
| | Gemma 4 31B | 0.816→0.850 | +0.034 (4.3) | 0.816→0.928 | +0.112 (10.2) |
| `running_sum_local` | 7B | 0.286→0.280 | −0.006 (0.8) | 0.277→0.348 | +0.071 (12.4) |
| | 32B | 0.547→0.603 | +0.056 (9.0) | 0.508→0.660 | +0.152 (12.1) |
| | Qwen3.8 27B | 0.441→0.573 | +0.131 (13.0) | 0.419→0.683 | +0.264 (19.4) |
| | Gemma 4 31B | 0.754→0.775 | +0.021 (3.2) | 0.723→0.840 | +0.117 (10.8) |
| `sliding_window_max` | 32B | 0.668→0.733 | +0.065 (7.9) | 0.677→0.816 | +0.139 (10.9) |
| | Qwen3.8 27B | 0.666→0.653 | −0.013 (1.0) | 0.640→0.824 | +0.183 (16.4) |
| | Gemma 4 31B | 0.708→0.783 | +0.075 (6.3) | 0.718→0.817 | +0.099 (7.9) |
| `copy_previous` | 7B | 0.508→0.418 | −0.090 (7.0) | 0.565→0.553 | −0.013 (1.4) |
| | 32B | 0.836→0.818 | −0.018 (1.9) | 0.874→0.792 | −0.083 (4.6) |
| | Qwen3.8 27B | 0.971→0.709 | −0.262 (23.9) | 0.942→0.836 | −0.105 (6.4) |
| | Gemma 4 31B | 0.917→0.832 | −0.086 (6.5) | 0.837→0.866 | +0.029 (1.3) |

## F External tasks

Tables 10 and 11 and Figures 10 and 11 give the results on tasks we did not design (Section 6) and on Gemini.

Table 10: External position-local tasks, pooled across length and density. Base: no-marker accuracy averaged over lengths. Best-$k$: accuracy at the single best spacing (optimistic). Δ: marker runs minus the length-matched no-marker mean (Section 3). ProcBench-Substitute was regenerated after we made the vendored ProcBench generator's substitution table deterministic (it was drawn in a process-dependent order).

| | Qwen2.5-7B | | | Qwen2.5-32B | | |
|---|---|---|---|---|---|---|
| Task | Base | Best-$k$ | Δ ($\|z\|$) | Base | Best-$k$ | Δ ($\|z\|$) |
| Vowel/consonant tagging | 0.438 | 0.435 | −0.083 (2.4) | 0.615 | 0.657 | +0.012 (0.5) |
| ProcBench-Substitute | 0.189 | 0.432 | +0.096 (6.7) | 0.390 | 0.757 | +0.203 (10.8) |
| String copying | 0.973 | 1.000 | +0.009 (1.0) | 0.993 | 1.000 | +0.007 (2.0) |

On Kadane's algorithm, base accuracy at both Qwen scales is statistically indistinguishable from chance under a shuffle test, so we do not interpret its effects. Minimum is well above chance at Qwen2.5-32B ($z = 4.1$ from shuffled chance).

Table 11: Closed-model check and CLRS-Text. Gemini 3.5 Flash with 3 trials per cell, pooled over all tested lengths (including $N$ up to 256 for `cyclic_shift`), with the no-marker mean length-matched. CLRS-Text tasks use token-level markers and scoring.

| Task | Model | Base TAR | Bounds TAR | $\Delta$ ($\lvert z \rvert$) |
|---|---|---|---|---|
| `cyclic_shift` | Gemini | 0.596 | 0.852 | +0.256 (14.4) |
| `random_permutation` | Gemini | 0.561 | 0.846 | +0.284 (8.4) |
| `conditional_shift` | Gemini | 0.453 | 0.628 | +0.175 (3.1) |
| `running_xor` | Gemini | 0.882 | 0.913 | +0.032 (0.4) |
| `running_sum_local` | Gemini | 0.918 | 0.803 | −0.116 (5.2) |
| `copy_previous` | Gemini | 0.984 | 0.924 | −0.060 (2.8) |
| CLRS minimum | Qwen2.5-7B | 0.133 | 0.183 | +0.050 (3.3) |
| | Qwen2.5-32B | 0.592 | 0.392 | −0.200 (4.5) |
| | Gemini | 0.997 | 0.654 | −0.343 (7.3) |
| CLRS Kadane | Qwen2.5-7B | 0.060 | 0.068 | +0.008 (1.0) |
| | Qwen2.5-32B | 0.186 | 0.122 | −0.064 (2.0) |
| | Gemini | 0.384 | 0.354 | −0.030 (0.3) |

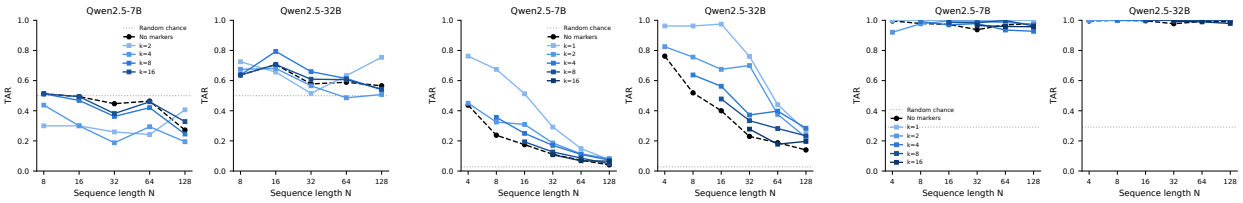



Figure 10: External tasks by marker density. Left: vowel/consonant tagging. Middle: ProcBench-Substitute. Right: string copying.

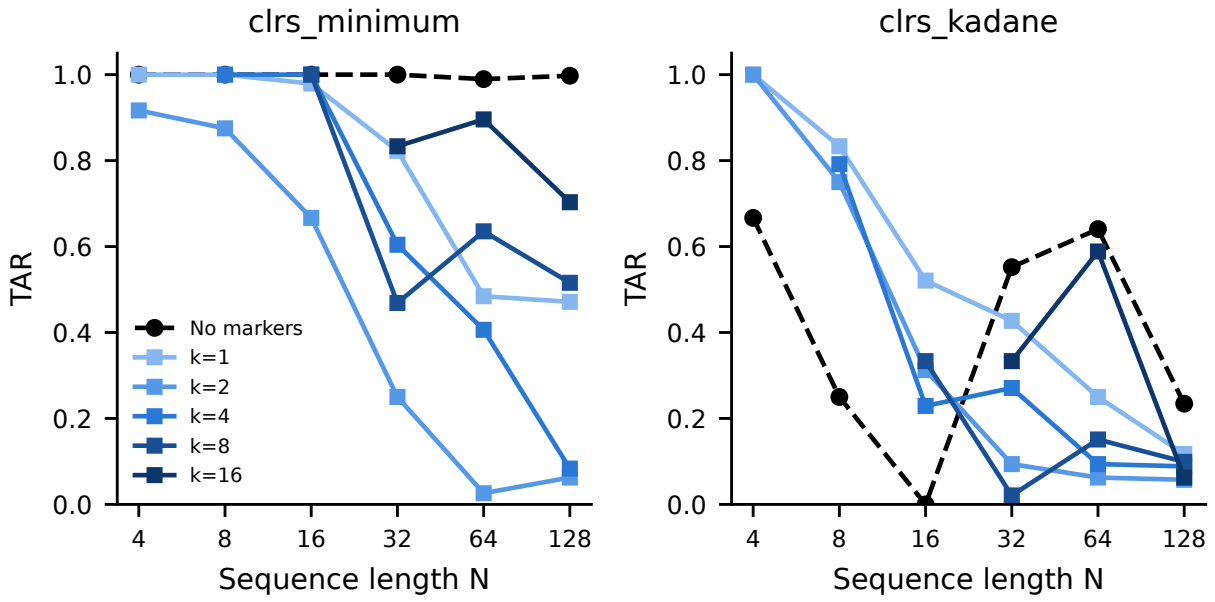



Figure 11: CLRS-Text tasks on Gemini 3.5 Flash by sequence length, one line per marker spacing $k$ (3 trials per cell). Dashed: no markers.

## G Mechanism details

This appendix collects the attention analyses behind Section 7: the mechanism overview (Figure 12), attention statistics by layer (Figure 13), example heads (Figure 14), the earlier head ablation (Figure 15), a digit-tokenization check of the attention pattern (Figure 16), error recovery after markers (Figure 17), the head taxonomy (Table 13), and the 7B ablation (Table 12). In Figure 17, the 50% rate away from markers is chance for this binary alphabet.

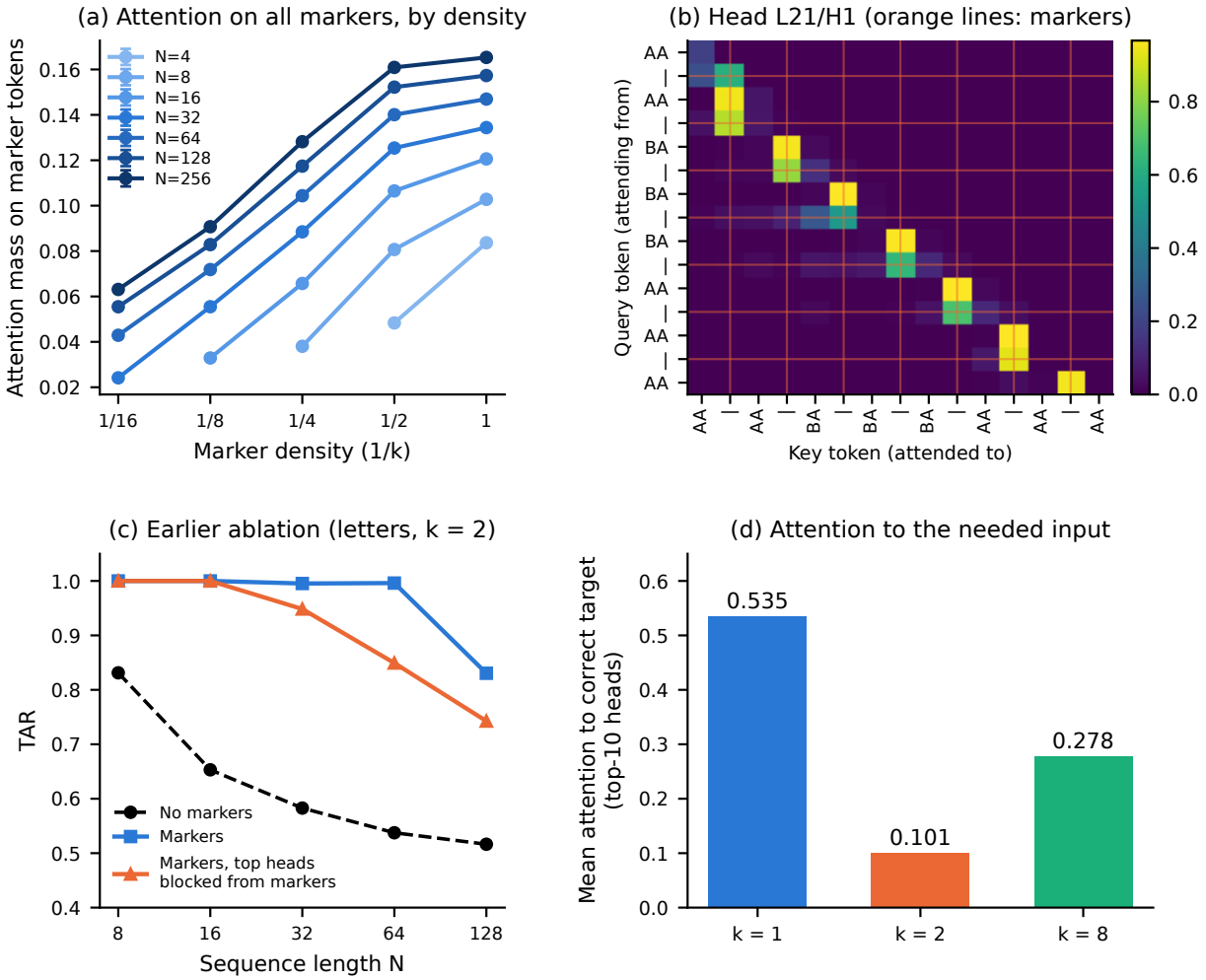



Figure 12: Mechanism (Qwen2.5-32B). (a) Total attention mass on all marker tokens rises with density at every length (`cyclic_shift`, $A = 2$); the number of markers rises faster, so this is not evidence that individual markers draw more attention than content tokens. (b) The most marker-attending head (layer 21, head 1) attends from each position to an adjacent marker. (c) Earlier ablation on contiguous letters ($k = 2$; Section 7): masking the ten most marker-attending heads' attention to markers removes a growing share of the benefit as $N$ grows (0% at $N \leq 16$, 11% at 32, 32% at 64, 28% at 128). (d) `copy_previous` at $N = 32$: mean attention from each output position to the input token it needs (top-10 heads) is 0.535 at $k = 1$, 0.101 at $k = 2$, and 0.278 at $k = 8$, matching accuracy (Base 0.83; $k = 1$: 0.88; $k = 2$: 0.65; $k = 8$: 0.84).

Table 12: Head ablation at Qwen2.5-7B, same design as Table 3. Marker gains are small at this scale, so shares are noisy; "–" marks cells with a gain below 0.03.

| | | | | Markers, with heads blocked | | | Share of gain removed | |
|---|---|---|---|---|---|---|---|---|
| Format | $N$ | No markers | Markers | Top 10 | Random 10 | Top 10, sham | Top 10 | Controls |
| Letters | 32 | 0.17 | 0.24 | 0.23 | 0.24 | 0.24 | 14% | 0% |
| Letters | 64 | 0.14 | 0.16 | 0.14 | 0.15 | 0.16 | – | – |
| Spaced letters | 32 | 0.42 | 0.55 | 0.51 | 0.52 | 0.55 | 26% | $\leq$ 22% |
| Spaced letters | 64 | 0.19 | 0.30 | 0.27 | 0.31 | 0.32 | 24% | 0% |
| Digits | 32 | 0.24 | 0.27 | 0.28 | 0.29 | 0.28 | – | – |
| Digits | 64 | 0.18 | 0.19 | 0.19 | 0.19 | 0.19 | – | – |

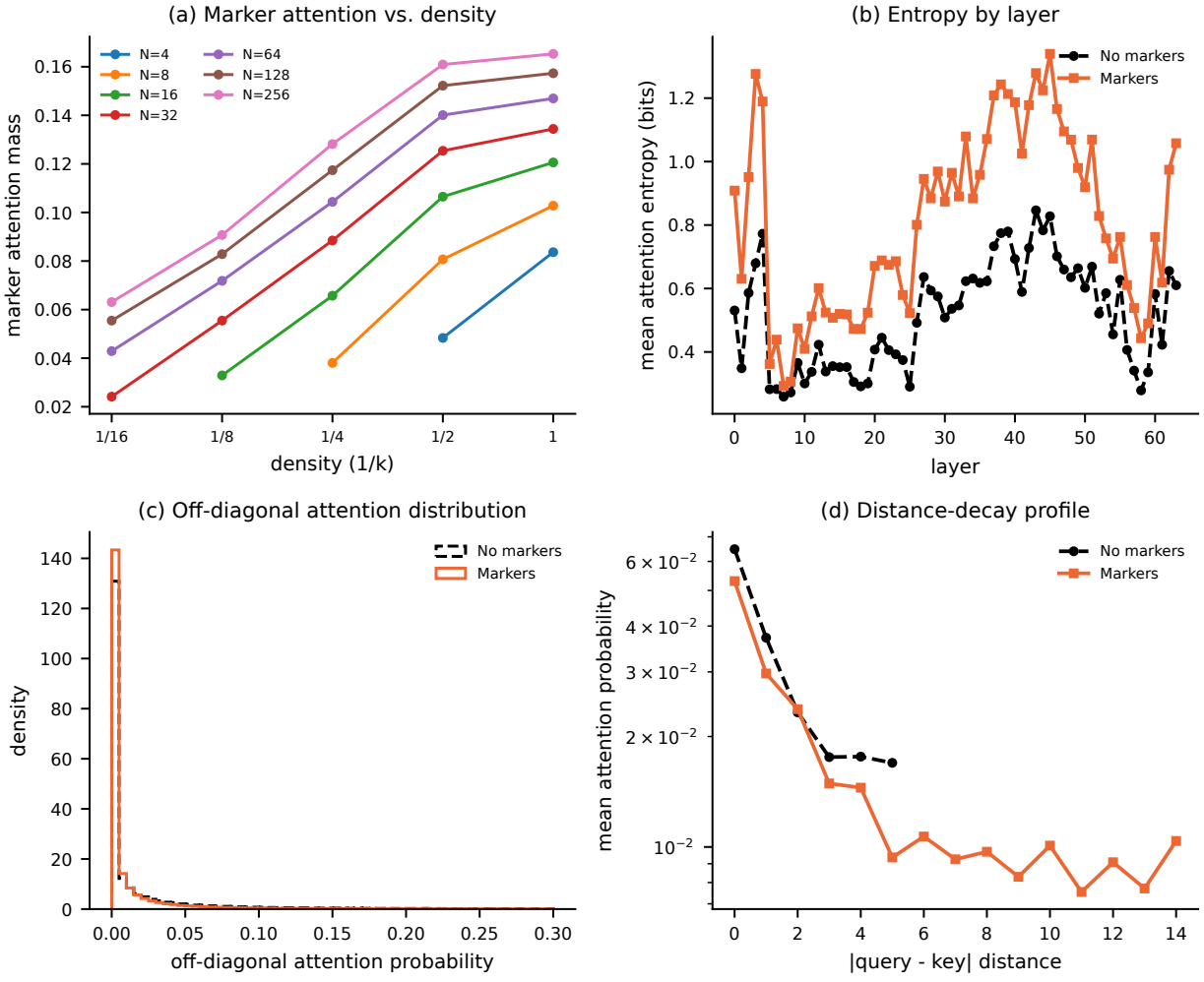



Figure 13: Attention analysis, Qwen2.5-32B, `cyclic_shift`: marker attention by density, attention entropy by layer (higher under markers at nearly every layer), off-diagonal attention, and distance decay.

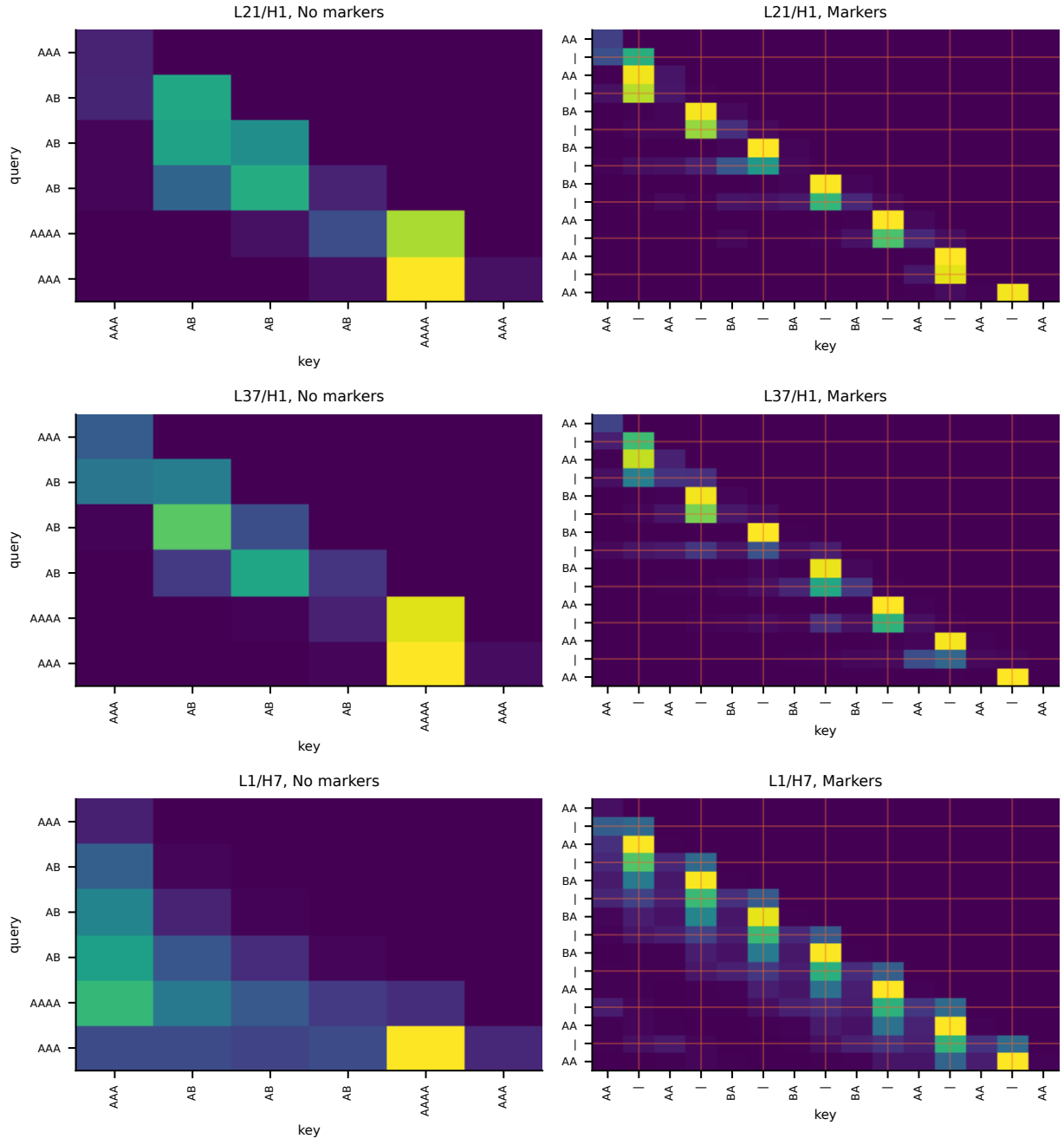



Figure 14: The three most marker-attending heads at Qwen2.5-32B (L21/H1, L37/H1, L1/H7). Without markers, letters merge into multi-character tokens; markers restore per-character resolution.

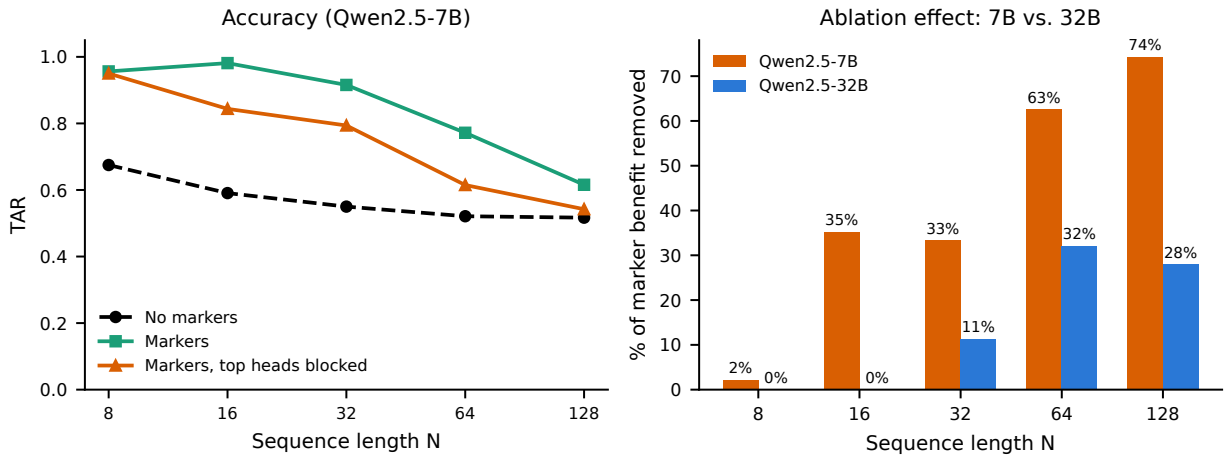



Figure 15: Earlier head ablation, contiguous letters (`cyclic_shift`, $A = 2$, $k = 2$), Qwen2.5-7B and 32B: share of the marker benefit removed by masking the ten most marker-attending heads' attention to markers, re-identified at each length. Unlike Table 3, Base and unablated runs come from an earlier grid and there are no controls at 32B.

Table 13: The ten most marker-attending heads at Qwen2.5-32B, with percentiles within all heads on an induction test and a positional (previous-token) test.

| Layer/Head | Marker attention | Induction pct. | Positional pct. |
|---|---|---|---|
| 21/1 | 0.099 | 38.3 | 98.3 |
| 37/1 | 0.087 | 47.4 | 96.7 |
| 1/7 | 0.086 | 17.1 | 97.7 |
| 35/19 | 0.082 | 23.1 | 96.5 |
| 35/3 | 0.080 | 14.7 | 97.2 |
| 30/32 | 0.079 | 18.6 | 96.0 |
| 32/37 | 0.075 | 14.3 | 98.8 |
| 1/8 | 0.072 | 64.8 | 97.7 |
| 32/7 | 0.071 | 60.4 | 90.2 |
| 19/19 | 0.068 | 9.9 | 83.9 |
| Mean | | 30.9 | 95.3 |

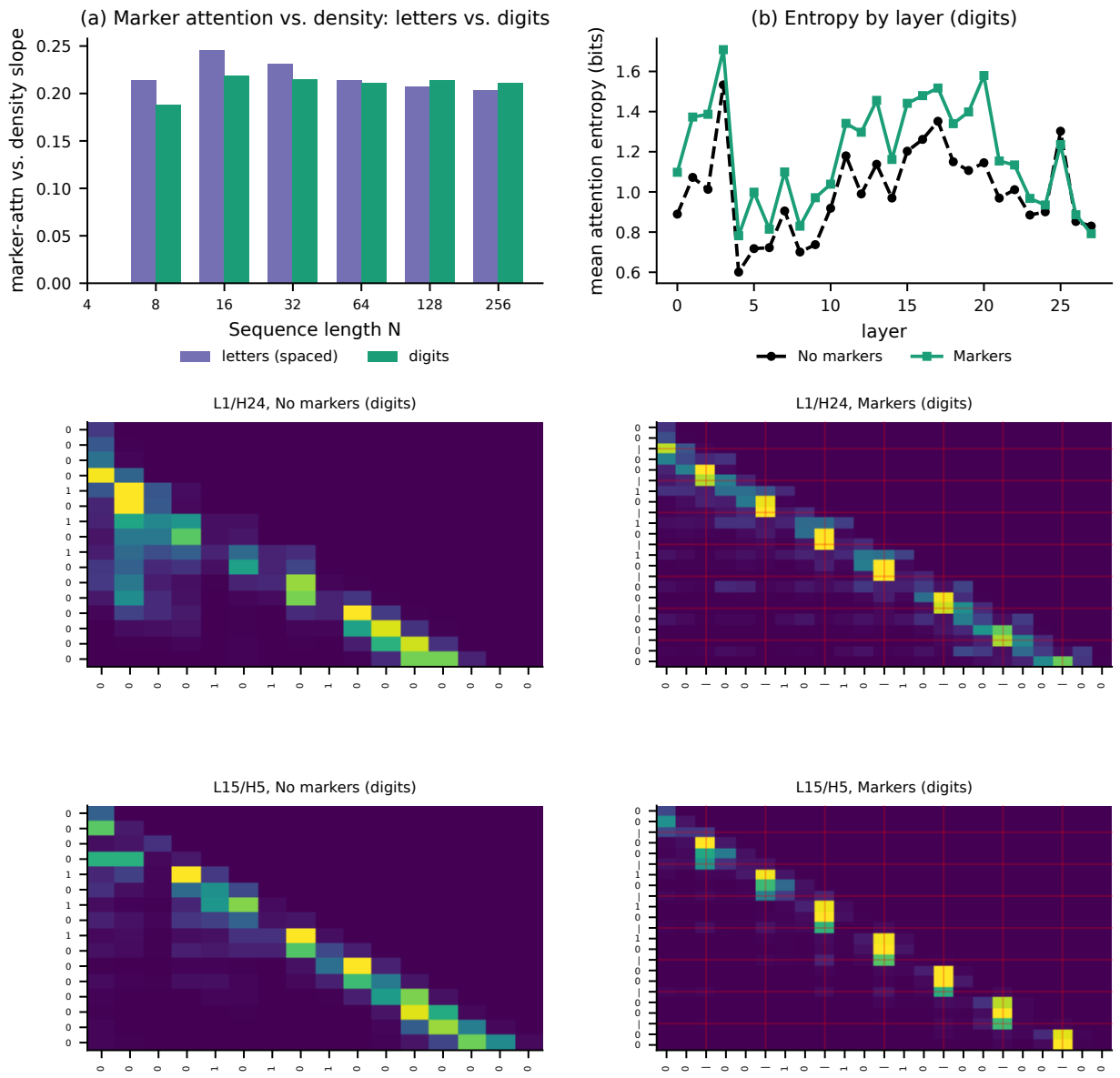



Figure 16: Tokenization check at Qwen2.5-7B: marker-attention slopes with letters (space-separated, one token per letter) and digits (natively one token per character) are nearly identical.

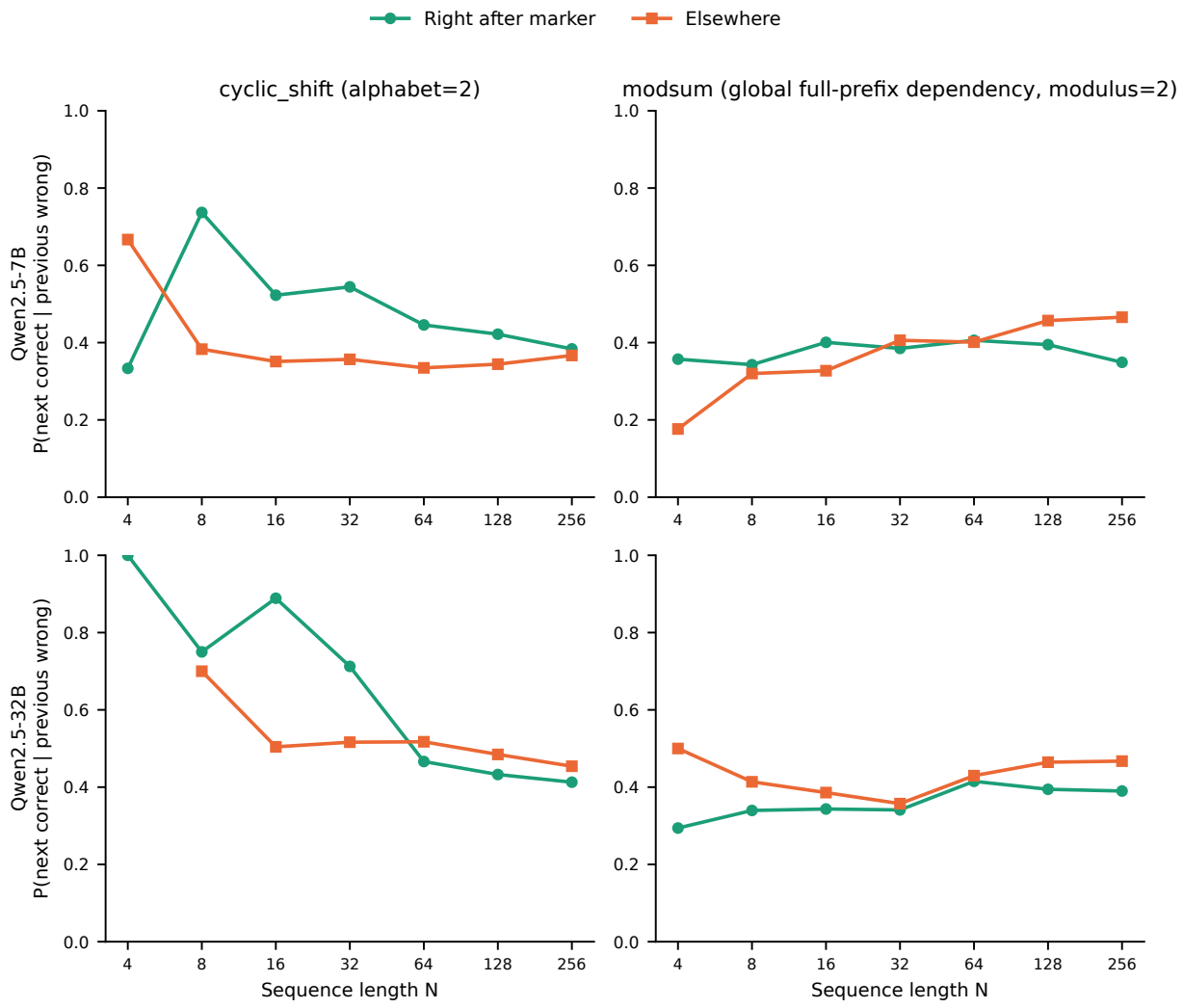



Figure 17: Error recovery: after a wrong output, the fraction of next positions that are correct, split by whether the next position follows a marker. Left: `cyclic_shift`; right: a full-prefix modular sum. $A = M = 2$.

## H Marker ablations

Tables 14–16 vary the marker symbol, its placement, and the prompt wording on contiguous-letter `cyclic_shift` (Section 8).

Table 14: Marker identity, reserved token, and token budget (`cyclic_shift`, $A = 2$, periodic $k = 4$ unless noted). TAR. Budget-matched: `<|>` at $k = 12$ uses the same total tokens as `|` at $k = 4$.

| $N$ | Model | Base | `\|` | `;` | `#` | Reserved token | `<\|>`, $k = 12$ |
|---|---|---|---|---|---|---|---|
| 8 | 7B | 0.675 | 0.717 | 0.792 | 0.792 | 0.469 | 0.675 |
| | 32B | 0.831 | 0.892 | 0.942 | 0.908 | 0.525 | 0.806 |
| 16 | 7B | 0.591 | 0.825 | 0.750 | 0.667 | 0.216 | 0.591 |
| | 32B | 0.653 | 0.896 | 0.846 | 0.912 | 0.284 | 0.759 |
| 32 | 7B | 0.550 | 0.679 | 0.769 | 0.585 | 0.092 | 0.539 |
| | 32B | 0.583 | 0.848 | 0.781 | 0.744 | 0.186 | 0.688 |
| 64 | 7B | 0.521 | 0.616 | 0.650 | 0.591 | 0.049 | 0.453 |
| | 32B | 0.537 | 0.759 | 0.658 | 0.693 | 0.160 | 0.710 |
| 128 | 7B | 0.517 | 0.545 | 0.507 | 0.496 | 0.046 | 0.452 |
| | 32B | 0.516 | 0.694 | 0.744 | 0.561 | 0.219 | 0.642 |

Table 15: Placement regularity (`cyclic_shift`, $A = 2$, marker count matched to $k = 4$). TAR.

| | Qwen2.5-7B | | | | | Qwen2.5-32B | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $N$ | Base | Periodic | $\pm 1$ | $\pm 2$ | Random | Base | Periodic | $\pm 1$ | $\pm 2$ | Random |
| 8 | 0.675 | 0.717 | 0.581 | 0.675 | 0.633 | 0.831 | 0.892 | 0.869 | 0.912 | 0.858 |
| 16 | 0.591 | 0.825 | 0.572 | 0.566 | 0.583 | 0.653 | 0.896 | 0.863 | 0.787 | 0.733 |
| 32 | 0.550 | 0.679 | 0.572 | 0.509 | 0.552 | 0.583 | 0.848 | 0.847 | 0.745 | 0.779 |
| 64 | 0.521 | 0.616 | 0.502 | 0.491 | 0.491 | 0.537 | 0.759 | 0.731 | 0.732 | 0.682 |
| 128 | 0.517 | 0.545 | 0.493 | 0.471 | 0.452 | 0.516 | 0.694 | 0.691 | 0.695 | 0.583 |

Table 16: Prompt wording (`cyclic_shift`, $A = 2$, $k = 4$, 15 trials per cell). $\Delta$ = Bounds − Base TAR.

| | Qwen2.5-7B, $N =$ | | | | | Qwen2.5-32B, $N =$ | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Wording | 8 | 16 | 32 | 64 | 128 | 8 | 16 | 32 | 64 | 128 |
| Formal | +.09 | +.27 | +.10 | +.13 | +.07 | +.26 | +.17 | +.17 | +.20 | +.18 |
| Casual | −.04 | +.08 | +.12 | +.04 | +.10 | +.08 | +.25 | +.17 | +.21 | +.20 |
| Terse cipher | −.07 | +.08 | +.00 | +.00 | +.05 | −.12 | −.11 | +.01 | −.07 | −.05 |

## I Delimiters, longer sequences, and sampling

Table 17 gives the follow-up experiment with commas and line breaks as markers and sequences up to $N = 256$, and Table 18 repeats three tasks with sampling instead of greedy decoding (Sections 5 and 6).

Table 17: Delimiter type and longer sequences at Qwen2.5-32B ($k \in \{4, 16\}$ pooled, 20 trials per cell; `random_permutation` with digits at $A = 8$, `running_sum_local` at modulus 4). Change in accuracy: marker run minus no-marker run, pooled over positions. Post-marker and within-segment: marker run minus the no-marker run at the same positions. Each delimiter is named in its own constraint sentence. The `|` cells at $N = 64$ reproduce the main grid exactly (same prompts, greedy decoding). No output reached the generation budget.

| | | Change in accuracy | | | Post-marker | | | Within-segment | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Task | $N$ | `\|` | , | line break | `\|` | , | line break | `\|` | , | line break |
| `random_permutation (digits)` | 64 | +0.52 | +0.52 | +0.53 | +0.56 | +0.56 | +0.53 | +0.52 | +0.52 | +0.53 |
| | 128 | +0.59 | +0.57 | +0.59 | +0.67 | +0.64 | +0.65 | +0.58 | +0.56 | +0.58 |
| | 256 | +0.46 | +0.46 | +0.47 | +0.53 | +0.53 | +0.50 | +0.46 | +0.45 | +0.46 |
| `running_xor` | 64 | +0.15 | +0.16 | +0.17 | −0.13 | −0.11 | −0.14 | +0.20 | +0.21 | +0.24 |
| | 128 | +0.18 | +0.13 | +0.21 | −0.03 | −0.05 | −0.04 | +0.22 | +0.16 | +0.26 |
| | 256 | +0.17 | +0.15 | +0.20 | +0.04 | +0.03 | +0.05 | +0.19 | +0.17 | +0.23 |
| `running_sum_local` | 64 | +0.13 | +0.11 | +0.15 | −0.13 | −0.11 | −0.12 | +0.17 | +0.15 | +0.19 |
| | 128 | +0.11 | +0.09 | +0.14 | +0.02 | +0.00 | +0.02 | +0.14 | +0.11 | +0.17 |
| | 256 | +0.08 | +0.07 | +0.12 | +0.03 | +0.02 | +0.05 | +0.09 | +0.09 | +0.14 |

Table 18: Sampling instead of greedy decoding at Qwen2.5-32B ($N \in \{32, 64\}$, $k \in \{4, 16\}$ pooled, 20 prompts per cell, one sample each; `random_permutation` at $A = 8$, `running_sum_local` at modulus 4). The greedy rows are the main-grid runs on the same prompts. Change: marker run minus the no-marker run of the same length and trial, with a 95% interval. Post-marker and within-segment: marker run minus the no-marker run of the same decoding regime at the same positions.

| Task | Decoding | No markers | Markers | Change | Post-marker | Within-segment |
|---|---|---|---|---|---|---|
| `random_permutation (digits)` | Greedy | 0.55 | 0.92 | $+0.37 \pm 0.05$ | — | — |
| | $T = 0.7$, top-$p$ 0.8 | 0.54 | 0.92 | $+0.38 \pm 0.05$ | — | — |
| | $T = 1.0$, top-$p$ 0.95 | 0.52 | 0.93 | $+0.42 \pm 0.05$ | — | — |
| `running_xor` | Greedy | 0.66 | 0.77 | $+0.11 \pm 0.03$ | −0.18 | +0.16 |
| | $T = 0.7$, top-$p$ 0.8 | 0.67 | 0.76 | $+0.09 \pm 0.03$ | −0.16 | +0.14 |
| | $T = 1.0$, top-$p$ 0.95 | 0.66 | 0.76 | $+0.10 \pm 0.03$ | −0.17 | +0.15 |
| `running_sum_local` | Greedy | 0.39 | 0.48 | $+0.09 \pm 0.03$ | −0.04 | +0.12 |
| | $T = 0.7$, top-$p$ 0.8 | 0.39 | 0.47 | $+0.08 \pm 0.03$ | −0.06 | +0.12 |
| | $T = 1.0$, top-$p$ 0.95 | 0.39 | 0.47 | $+0.08 \pm 0.03$ | −0.07 | +0.11 |